LED Blink Circuit

Arduino Uno R3 · Tinkercad Simulation

A beginner Arduino project that blinks a single LED ON and OFF every second, using digital pin 4 controlled entirely through code (pinMode, digitalWrite, delay).

Components Used

Component	Quantity	Purpose
Arduino Uno R3	1	Microcontroller board — runs the blink code
LED (any color)	1	Visual output — turns ON and OFF
Resistor (220Ω)	1	Limits current to protect the LED
Jumper Wires	2	Connect pin 4 to LED, and LED/resistor to GND
Breadboard (optional)	1	Not required in Tinkercad if using direct wiring

Circuit Diagram

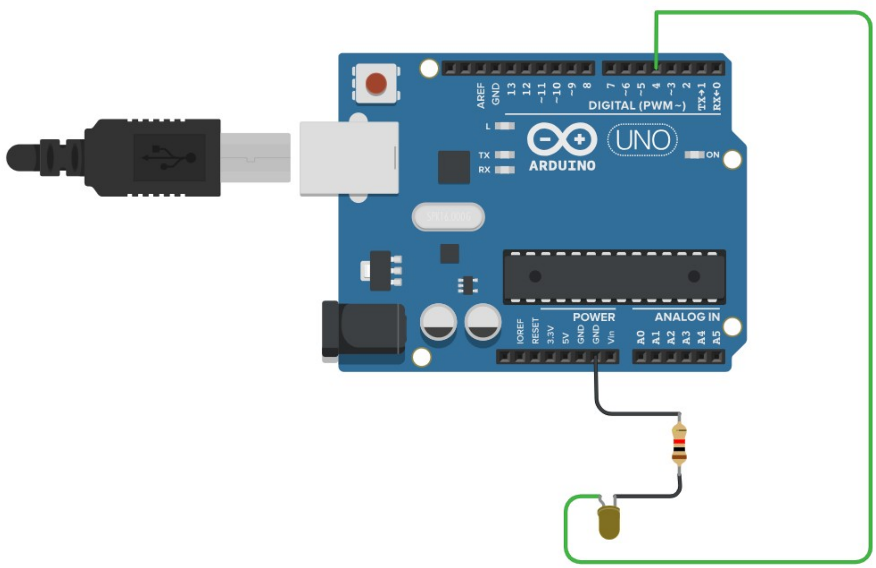

Fig 1: LED Blink circuit in Tinkercad — pin 4 (green wire) to LED anode, LED cathode through 220Ω resistor to GND.

Build Steps

1. **Open Tinkercad —** Go to tinkercad.com, sign in, and create a New Circuit.
2. **Add Arduino Uno R3 —** Drag an Arduino Uno R3 board from the components panel onto the workspace.
3. **Add the LED —** Drag an LED onto the workspace and place it near the Arduino.
4. **Add the Resistor —** Drag a resistor next to the LED and set its value to 220Ω (click the resistor to edit).

5. **Wire Pin 4 to LED Anode —** Connect a wire from Arduino digital pin 4 to the LED's anode (longer leg, +).

6. **Wire LED Cathode to Resistor —** Connect the LED's cathode (shorter leg, -) to one leg of the 220Ω resistor.

7. **Wire Resistor to GND —** Connect the other leg of the resistor to a GND pin on the Arduino.

8. **Add the Code —** Click the Code button, switch to Text mode, and paste the sketch below.

9. **Run the Simulation —** Click Start Simulation — the LED should blink ON for 1 second and OFF for 1 second, repeatedly.

Arduino Code

```
void setup()
{
  pinMode(4, OUTPUT);
}

void loop()
{
  digitalWrite(4, HIGH);
  delay(1000); // Wait for 1000 millisecond(s)
  digitalWrite(4, LOW);
  delay(1000); // Wait for 1000 millisecond(s)
}
```

Wiring Summary

From	To	Wire Color
Arduino Pin 4	LED Anode (long leg / +)	Green
LED Cathode (short leg / -)	Resistor (220Ω) — one leg	Black
Resistor (220Ω) — other leg	Arduino GND	Black

Kaise Kaam Karta Hai

Ye circuit ek simple ON/OFF logic pe kaam karta hai. Arduino ka pin 4 digitalWrite() function se HIGH aur LOW hota rehta hai — jab pin HIGH hota hai to us se 5V current LED ki anode (positive leg) mein jata hai, LED ON ho jati hai.

Resistor (220Ω) LED ke cathode side pe lagaya gaya hai taake current ki value limit ho aur LED jal na jaye (bina resistor ke bohat zyada current flow hota hai jo LED ko damage kar sakta hai).

delay(1000) function 1000 milliseconds (1 second) tak wait karta hai — is se pehle LED ko HIGH pe 1 second rehne dete hain, phir LOW kar ke 1 second OFF rakhte hain. Ye loop() function ke andar hamesha repeat hota rehta hai, is liye LED continuously blink karti rehti hai.

Important baat: LED ko seedha 5V pin se connect nahi karte, kyunke 5V se LED hamesha ON rahegi (constant), blink nahi karegi. Blink sirf tab hota hai jab hum ek digital pin (jaise pin 4) use karte hain aur code se us pin ko HIGH/LOW switch karte hain.